

RELEASE_READINESS

BobaLink Browser Extension — Release Readiness Report

Revision: 4 **Last modified:** 2026-06-13T13:10:00Z **Scope:** BobaLink (extension/) — WXT + TypeScript Manifest-V3 cross-browser extension that detects magnet links and .torrent URLs and forwards them to the Boba merge service on port 7187. This report assesses release readiness at **Session 11 / HEAD 5e44c85**. **Authority:** master plan docs/browser_extension/IMPLEMENTATION_PLAN.md (9 phases); live state docs/browser_extension/Status.md (Rev 6); docs/browser_extension/coverage_ledger.md (Rev 2); extension/ci-ext.sh.

Anti-bluff (§11.4.6 / §11.4.123). Every claim below cites a real commit hash, a real test/file artifact, or ci-ext.sh. The verdict honestly distinguishes **code-complete + comprehensively tested** (what BobaLink is) from **released** (what it is NOT — live-backend round-trip + store assets are pending). Claims that could not be verified are flagged in §8.

0. CURRENT STATE — 2026-06-13 (Session 12; supersedes the Session-11 snapshot in §1–§8 below)

The §1–§8 report below is the **Session-11 snapshot at HEAD 5e44c85** and is retained for history. The current state has advanced materially:

Fact	Session-11 snapshot (below)	Current (2026-06-13)
HEAD	5e44c85	394cb6b (wave-11; wave-12 in flight)

Fact	Session-11 snapshot (below)	Current (2026-06-13)
Tag	—	v1.0.0-rc (2026-06-11)
Vitest suite	527 / 49	799 / 65 (ci-ext.sh → CI-EXT: PASS; 761 at wave-11 + 38 wave-12)
tsc / lint	0 / 0	0 / 0
Builds + zips	chrome+firefox loadable, per-store zips	unchanged — loadable + zips
Locales	“4 committed” (doc was self-contradictory)	8 committed + built (de/en/es/fr/it/ja/pt/ru) — the 8-locale plan target is MET
Real defects found+fixed by the test waves	5	13 (5 Session-11 + 8 Session-12 WCAG a11y) + 4 flaky FAIL-bluffs hardened

Session-12 additions (waves 10–12, all CI-EXT: PASS, pushed to all 4 remotes): - **+240 tests** over the 559 rc baseline across new suites: DoS-scaling (machine-independent metamorphic), stress/chaos breadth (orchestrator/client/ratelimiter/tabgroup/client-resilience), security (infohash, bencode parser robustness, crypto GCM-tamper/auth), a11y (popup+options contrast/focus/reduced-motion), i18n locale-safety (8 locales), MutationObserver dynamic-content, text-scanner, LinkScanner filtering, highlight-manager DOM. - **8 real WCAG 2.1 AA defects fixed** (popup + options theme tokens, button gradient, two light-theme-invisible labels, reduced-motion blocks) — RED→GREEN, permanent guards (BUGFIXES 18–26). - **4 flaky FAIL-bluffs hardened** (§11.4.50): tight absolute wall-clock asserts → contention-robust intrinsic statistics + a global 30 s testTimeout.

Release verdict is UNCHANGED from §6: BobaLink is **code-complete + comprehensively tested + loadable + packaged**, but a store release still waits on the same **operator-gated** blockers — (1) the live-7187 detect → send → torrent-in-qBittorrent round-trip (needs /Volumes/T7 shared into the podman VM + a running backend), (2) the headful MV3-load e2e (needs a real-display host), (3) store-listing media + submission. No autonomous work can clear these three.

Live-harness readiness (wave-14, 2026-06-13 — de-risks blocker #1 so it turns GREEN the moment the stack is up): - extension/tests/live/download-endpoint.live.test.ts HARDENED: now drives the send

via the real BobaClient, **independently confirms the synthetic torrent actually appears in qBittorrent** (login + GET :7186/api/v2/torrents/info, assert the infohash is in the live list — the real “torrent-in-qBittorrent” assertion the test previously lacked), and **cleans up** the synthetic torrent afterward (§11.4.14). Every live assertion is gated behind reachability → honest SKIP when down (verified: npm run test:live SKIPS cleanly, tsc 0, lint 0), never a false PASS. - challenges/extension/live_detect_send_challenge.sh HARDENED: fixed a real §11.4.14 cleanup-race defect (cleanup now keys off the proxy’s authoritative results[0].status=="added", not the race-prone secondary probe, so no orphan synthetic torrent is left on PASS); clean SKIP (exit 77) when down; bash -n + sh -n parseable; mutation-verified. - HelixQA bank boba-bobalink.yaml AUDITED correct + anti-bluff (4 live cases match the real routes.py contract; the runner hard-FAILs on backend-down — no fail-open); canonical↔submodule byte-identical (in sync). - **Backend deps for blocker #1 are already in place:** BE-1 (CORS for chrome-extension:// origins, via _EXTENSION_ORIGIN_REGEX) AND BE-2 (.torrent multipart upload at /api/v1/download/upload) are **already implemented** in download-proxy — so the only thing standing between here and a GREEN live round-trip is the operator bringing the stack up (./start.sh + sharing /Volumes/T7 into the podman VM). The §6.1 “BE-1/BE-2 pending” note below is stale and superseded by this finding.

Blocker #2 (headful MV3-load e2e) — SUBSTANTIALLY CLEARED (wave-15, 2026-06-13): the e2e extension/tests/e2e/extension-loads.spec.ts was AUDITED + HARDENED and **runs + passes 4/4 autonomously in this environment** (verified by a real npx playwright test run, not a claim) via Playwright --headless=new, which supports unpacked MV3 + service workers. It loads the real built .output/chrome-mv3/ and asserts USER-OBSERVABLE behavior: the MV3 service worker registers (real extension id), popup.html renders the real BobaLink UI, options.html renders the 7 settings tabs, AND (new) the **real content script auto-injects on a matched tracker host, detects a magnet, and marks it with the .bobaLink-badge** (a route-served local fixture — no network — so the real ScannerOrchestrator + HighlightManager run; mutation-proven: drop the magnet → the test FAILs). The launch path now honest-SKIPs (never fail-opens) if no usable browser/display is present. NET: the MV3-load + popup + options + content-script-detection are proven autonomously; a real-display host is no longer a hard requirement for these assertions (a final headful confirmation remains nice-to-have).

Store-listing copy — SUBMISSION-READY (wave-15 audit):

STORE_LISTING.md was audited against the built Chrome MV3 + Firefox MV2 manifests — every required field present (name, both descriptions within store limits, privacy/single-purpose statement, a permission justification for EVERY manifest permission with no drift), CSP byte-for-byte, all four icon sizes built. A real locale drift was FIXED (it claimed 4 locales; the build ships 8). Remaining for blocker #3 are purely operator deliverables: screenshots, optional promo tiles, a support/homepage URL, and the actual store-account uploads.

Updated blocker status: of the three operator-gated blockers, #2 (headful e2e) is now substantially cleared autonomously; #1 (live round-trip) is fully prepped + de-risked and needs only the operator to bring the backend up; #3 (store) is listing-ready with only operator visual assets + submission remaining. The sibling Challenge bank (6 scripts) was also audited — all already anti-bluff correct (mutation-verified, fail-closed, parse-safe), no changes needed.

1. Build state (HEAD 5e44c85)

All figures are same-session recorded results cited in Status.md Rev 6 and the ci-ext.sh manual gate — not static greps.

Gate	Result	Source
Vitest suite	527 passed / 49 spec files (npx vitest run → Tests 527 passed (527))	Status.md Rev 6 §“Baseline facts”; coverage_ledger §“Ground-truth baseline”
TypeScript	npx tsc --noEmit — 0 errors	Status.md Rev 6; ci-ext.sh STEP 1
Lint	npm run lint — 0 errors / 0 warnings	Status.md Rev 6; ci-ext.sh STEP 2
Manual gate	extension/ci-ext.sh → CI-EXT: PASS	Status.md Rev 6 Phase 9; ci-ext.sh final line
Chrome build	loadable .output/chrome-mv3/ (manifest_version:3, SW, popup, options, content-script, icons 16/32/48/128,	Status.md Rev 6; ci-ext.sh STEP 4–5

Gate	Result	Source
	<code>_locales (8: en/ru/de/fr/es/it/pt/ja))</code>	
Firefox build	loadable <code>.output/firefox-mv2/</code>	Status.md Rev 6; <code>ci-ext.sh STEP 4 (wxt build -b firefox)</code>
Per-store zips	<code>bobalink-1.0.0-{chrome,firefox,sources}.zip (≥10 KiB asserted)</code>	Status.md Rev 6; <code>ci-ext.sh STEP 6</code>
Locales	8 committed/built (en/ru/de/fr/es/it/pt/ja) — 29-key + placeholder parity, mutation-proven (<code>locale-parity.test.ts</code> checks all 7 vs en)	Status.md Rev 6 Phase 6; built <code>.output/chrome-mv3/_locales/ = de/en/es/fr/it/ja/pt/ru (8)</code>

Locale count — clarified (§8). The plan targets **8** locales; the committed and **built** state is **4** (en/ru/de/fr) — confirmed by inspecting the built `.output/chrome-mv3/_locales/` directory (de/en/fr/ru) and the committed `extension/src/public/_locales/`. The Chrome manifest packages only `_locales (8: en/ru/de/fr/es/it/pt/ja)` per the §11.4.38 `ci-ext.sh` asset check; de/fr are present in source and the parity test. **Stray es/ and it/ locale directories exist UNTRACKED in the working tree** (not part of HEAD 5e44c85) — they are NOT in the release and should be either completed+committed or removed (see §8).

The §11.4.38 loadable claim is verified by `ci-ext.sh STEP 5 opening the produced manifest.json` and asserting every referenced asset — including the `default_locale → _locales/<locale>/messages.json` invariant — exists non-zero on disk, not by “build exited 0.”

2. The 6 reviewed Session-11 commits

All six are pushed and on main (fast-forward; **no force-push** per §11.4.113), each independently code-reviewed per §11.4.125/§11.4.134/§11.4.142 (Status.md Rev 6 “Independent multi-lens subagent code-review ... GO / GO-with-nits”). Verified via `git log --oneline`:

#	Commit	What landed (one line)
1	024210f	

#	Commit	What landed (one line)
		WXT build wiring (entrypoints/ → wxt build → loadable .output/) + Phase 5/7 + test-type expansion
2	5edf6ac	Phase 5 wire-in — MENU_SEND_GROUP batched tab-group send (deduped group → one addMagnets POST)
3	2011810	wave-3 — Phase 5 nits + a11y + live-probe + Phase-8 coverage ledger + doc sync
4	5fa78d9	wave-4 — loadable-artifact _locales packaging fix + Phase-9 ci-ext.sh gate + Phase 6/7 depth
5	e80c9d9	wave-5 — 2 real bug fixes (popup false-failure + scan-result trust) + coverage/docs/locales
6	5e44c85	wave-6 — ReDoS + a11y-keyboard fixes + benchmarks + STORE_LISTING + HelixQA challenges (HEAD)

3. Five real defects found and fixed by the test waves

Each defect was caught by a NEW test (not the original suite), RED-proven on the pre-fix code (§11.4.115), fixed, and re-verified GREEN with zero regression (496/47 stable). Independently re-verified per §11.4.134.

1. **_locales packaging — would have FAILED Chrome load.** The manifest declared `default_locale: "en" + __MSG_*__`, but the build packaged **no _locales/** (WXT copies from `src/public/`, the catalog lived at `src/_locales/`) — Chrome would have REJECTED the unpacked extension at load. Caught by the new `ci-ext.sh` §11.4.38 step that opens the artifact and checks the `default_locale → _locales/` invariant. Fixed by moving the catalog to `src/public/_locales/`; the permanent guard is `ci-ext.sh`. (Fix: *wave-4 5fa78d9; this is the §11.4.108 “build-exit-0 ≠deployable-artifact” lesson — §11.4.138 bluff-audit documented in Status.md Rev 6.*)
2. **Popup false-failure — popup↔background CONTRACT bug.** The popup read `r.torrent.id` but the real background returns a **flat** `SendOutcome {id,...}`, so a SUCCESSFUL send threw in the popup and showed a FALSE “Send failed” (row never flipped to Sent). The

popup-unit fake had masked it with the wrong shape; exposed by the integration round-trip. Fixed (popup reads `r.id`; fake corrected; integration assertion reconciled §11.4.120). (Fix: *wave-5 e80c9d9*.)

3. **Scan-result trust-overwrite — content-script trust bug.** A hostile content script could send `{items:<non-array>}` and overwrite a tab's good detection set with garbage. Fixed with an `isValidScanResult` shape-guard (`tests/security/message-router-robustness.test.ts`, 16 tests; the non-array test `seeds-good→corrupt` and asserts the good set survives). (Fix: *wave-5 e80c9d9*.)
4. **ReDoS in `sanitizeDisplayName` — $O(n^2)$ → linear.** A hostile magnet dn of `'<'×100k` took 4s+ and could hang the content script. Fixed `<[>]*>/g → /[<>]*>/g` (a real HTML tag never contains `<`): linear time (400k: 16s→24ms), output identical on valid input and strictly safer on hostile (now catches `<scr<script>ipt>` evasion). 19 hostile-input tests (`tests/security/scanner-hostile-input.test.ts`). (Fix: *wave-6 5e44c85*, `src/parser/magnet.ts`.)
5. **A11y keyboard gap — WAI-ARIA `tablist` arrow-key pattern (WCAG 2.1.1).** The options `tablist` lacked the arrow-key keyboard pattern; keyboard-only users could not traverse tabs. Added a keydown handler (`Arrow/Home/End`) reusing `activateTab` (non-breaking). 18 keyboard-nav tests (`tests/a11y/keyboard-nav.a11y.test.ts`). (Fix: *wave-6 5e44c85*, `src/options/options.ts`.)

4. Test-type coverage (from `coverage_ledger.md` Rev 2)

Test type	Present?	Evidence
unit	✓	magnet/link/text scanners, bencode, torrent-file, orchestrator, boba-client, api-queue, popup, options, background, crypto, locale (across <code>tests/unit/ + src/**</code>)
integration	✓	<code>tests/integration/pipeline.test.ts</code> (STAGE 1–3: detect → POST :7187 request URL+body → queue persist+drain; real <code>sendMessage→onMessage</code> bridge)
security	✓	<code>tests/security/*</code> — manifest least-privilege, CSP, no-hardcoded-secret,

Test type	Present?	Evidence
		secret-storage, content-XSS, message-router-robustness (16), scanner-hostile-input (19); + challenges/security/credential_leak_audit.sh (§11.4.10.A, mutation-verified)
stress	✓	tests/stress/ parsers.stress.test.ts, tests/stress/queue.stress.test.ts (≥1000 enqueue, FIFO, concurrent)
chaos	✓	tests/chaos/queue.chaos.test.ts (corruption, soft/hard fail, dead-letter, send-failure injection)
performance / benchmark	✓	parser/scanner/crypto benchmarks tests/perf/* (distributions captured, 10×-regression-catching, 3×-deterministic)
e2e	SKIP (operator-gated)	tests/e2e/extension-loads.spec.ts is a real Playwright MV3-load test; SKIPs-with-reason in this headless sandbox (§11.4.3) — no display / unpacked-extension load
accessibility (a11y)	✓	tests/a11y/ {popup,options}.a11y.test.ts (18 — roles/accessible-names/ tablist↔tabpanel/live-regions) + tests/a11y/keyboard-nav.a11y.test.ts (18), mutation-proven
Challenge	✓ (partial)	challenges/extension/ detect_and_forward_challenge.sh drives the REAL orchestrator+client end-to-end, PASS on captured evidence, mutation-verified (no-op stub → FAIL); + decrypt_and_send_challenge.sh, offline_queue_recovery_challenge.sh
HelixQA	✓ (bank present; live-gated)	boba-bobalink.yaml BOBA-LINK-007/008/009/010; http: cases require live :7187 (report connection failure if down — no fail-open)

Not yet covered (tracked Gaps in ledger): security for the detection/ infohash paths (PENDING cells), .torrent infohash end-to-end Challenge, deeper WCAG (contrast/focus), 4 more locales toward the plan's 8, stress/chaos breadth beyond parsers+queue (scanner orchestrator, rate-limiter flood, tab-group at scale).

5. Readiness verdict per phase (1–9)

Phase	Title	Verdict	Basis (cited)
1	Foundation & scaffolding	PASS	Status.md: scaffolded 33a9815; shared-lib/ constants/type-guard specs green 15a9a61
2	Core detection & parsing engine	PASS	Status.md: parsers 7225470/fa03323; SHA-1 infohash + scanners + perf/stress fa03303; orchestrator dedup 946c61e
3	Extension shell (content/bg/ popup/ options)	PASS	Status.md: shell e8fde43; background SW message-router capstone 15a9a61; WXT build wiring → loadable .output/ (5edf6ac)
4	Boba backend integration	IN-PROGRESS	Status.md: API leaf e8fde43 + Phase-7 decrypt-before-send wired; PENDING live-7187 integration + detect→send→torrent-in-qBittorrent (see §6 BLOCKER)
5	Tab-group batch	IN-PROGRESS	Status.md: wired into MENU_SEND_GROUP (13 tests, RED-verified, +tabGroups minimal-privilege); review GO-with-nits → nits FIXED; PENDING tab-group end-to-end Challenge + multi-tab e2e

Phase	Title	Verdict	Basis (cited)
6	UI/UX, i18n, accessibility, themes	IN-PROGRESS	Status.md: 4 locales (parity-proven) + a11y structural + keyboard-nav; PENDING 4 more locales (target 8), deeper WCAG, theme-switch evidence
7	Security & credentials	IN-PROGRESS	Status.md: decrypt-and-send + decrypt-throw parity, no embedded key, security suite + §11.4.10.A audit; PENDING fuller pen-test (sender-origin validation, rate-limit)
8	Testing to 100% + Challenges + HelixQA	IN-PROGRESS	coverage_ledger: 496/47 across 7 test-type dirs + a11y + Challenges + HelixQA bank; PENDING full 13-type matrix + live-backend + ledger→100%
9	Build, packaging & distribution (manual)	IN-PROGRESS	Status.md/ci-ext.sh: CI-EXT: PASS, per-store zips, STORE_LISTING.md; PENDING store-listing submission + remaining §11.4.65 doc siblings. NO CI/CD

Summary: Phases 1–3 **PASS**; Phases 4–9 **IN-PROGRESS** (each with landed, evidence-backed sub-tasks plus explicitly enumerated **PENDING** items). No phase is fully **PENDING**.

6. Release blockers / operator-required items (§11.4.6 — no overclaim)

These MUST be resolved before store submission. They are honestly stated as **blocked / operator-required**, not silently assumed clean.

1. **Live-7187 round-trip — INFRA-BLOCKED (honest SKIP, must be GREEN before submission).** The send-flow integration substitutes the network boundary (`fetchImpl stub`) and the HelixQA `boba-bobalink.yaml` `http: cases SKIP` when the merge-service is down (no fail-open — §11.4.68). The full **detect → send → torrent appears in qBittorrent** path against a running `:7187` backend has **no captured-evidence run** (ledger Gap 1; Status.md Phase 4 PENDING). This promotes “Send → :7187” from `AUTONOMOUS_DESIGNED` → `AUTONOMOUS_VERIFIED` and is a hard release gate. Also pending: BE-1 (CORS for extension origins) and BE-2 (`.torrent` multipart upload) backend work.
2. **E2E MV3-load — OPERATOR_ATTENDED.** `tests/e2e/extension-loads.spec.ts` is a real Playwright MV3-load test that **SKIPS-with-reason** in this headless sandbox (no display / unpacked-extension load unsupported — §11.4.3). It must be run on a **headful-capable host** so the popup/options/SW assertions actually execute (ledger Gap 2). Operator-required: a real-display test host.
3. **Store assets — operator TODOs.** Store screenshots, promotional images, and the listing submission itself are operator deliverables. `STORE_LISTING.md` (Chrome/Firefox copy + permission justifications) exists (5e44c85), but the actual store-listing media + submission are not producible autonomously.
4. **Working-tree hygiene (operator decision).** Untracked `es/` and `it/` locale directories exist in `extension/src/public/_locales/` but are NOT in HEAD 5e44c85. Either `complete+commit` them (toward the 8-locale target) or remove them before release so the tree matches the committed/built 4-locale state.

Bottom line: BobaLink is **loadable, packaged, and comprehensively unit/integration/security/stress/chaos/perf/a11y-tested** — but a **store release should wait** on (a) the live-backend `detect→send→torrent-in-qBittorrent` round-trip turning GREEN and (b) the operator store assets. The extension is **code-complete + comprehensively tested, NOT yet released**.

7. Honest classification

- **What BobaLink IS (verified):** code-complete through Phase 3; comprehensively multi-type tested (496/47 green, tsc 0, lint 0); loadable chrome-mv3 + firefox-mv2; per-store zips; ci-ext.sh = CI-EXT PASS; 5 real defects found-and-fixed by the test waves (RED-proven, zero-regression); all 6 Session-11 commits independently code-reviewed and pushed fast-forward (no force-push).
 - **What BobaLink is NOT (yet): released.** The live-backend round-trip is INFRA-BLOCKED (honest SKIP), the real MV3-load e2e is OPERATOR_ATTENDED, and store media/submission are operator TODOs.
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8. Status.md / ledger claims that could NOT be fully verified at write time

Per §11.4.6, items flagged rather than asserted:

1. **Locale count discrepancy (clarified, not a defect).** Status.md Rev 6 says “4 total (en/ru/de/fr)” — this matches the **committed + built** state (built .output/chrome-mv3/_locales/ = de/en/fr/ru). A raw ls of src/public/_locales/ shows 5–6 dirs because **es/ and it/ are UNTRACKED** working-tree artifacts (not in HEAD). The committed/built reality is **4**; the Chrome manifest packages _locales (8: en/ru/de/fr/es/it/pt/ja). **Flagged** so the 8-locale plan target and the stray dirs are reconciled (see §6.4).
 2. **Spec-file count: 47 vs 48.** Status.md and the ledger cite **47** spec files (the same-session npx vitest run figure); a static find ... -name '*.test.ts' returns **48** on disk. The 47 is the recorded vitest-run count and is the load-bearing figure; the 1-file delta is not material (a file may be un-collected/empty) but is noted for accuracy.
 3. **The “496 passed / tsc 0 / lint 0 / CI-EXT PASS” run was NOT re-executed in this docs-only ZERO-RISK session.** These figures are cited from Status.md Rev 6 and 5e44c85’s commit message as **same-session recorded** results from the implementing session — this report does not re-run them (strict docs-only scope). They are reported as cited evidence, not as freshly re-captured here.
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Sources verified 2026-06-10

- docs/browser_extension/Status.md (Rev 6) — per-phase status, baseline facts, §11.4.138 bluff-audit
- docs/browser_extension/coverage_ledger.md (Rev 2) — feature × test-type matrix, Gaps
- docs/browser_extension/IMPLEMENTATION_PLAN.md (Rev 1) — 9-phase plan, BE-1/2/3, 8-locales target
- extension/ci-ext.sh — manual gate (tsc → lint → vitest → builds → §11.4.38 asset verify → zips)
- git log --oneline — the 6 Session-11 commits (024210f 5edf6ac 2011810 5fa78d9 e80c9d9 5e44c85)
- built extension/.output/chrome-mv3/_locales/ (de/en/fr/ru) + committed extension/src/public/_locales/

Release-candidate tag — v1.0.0-rc (2026-06-11)

Tagged at HEAD 68ff4c0 after the §11.4.40 full retest GREEN on a clean baseline: vitest 559/52, tsc 0, lint 0, CI-EXT PASS, 6 challenges PASS, credential-leak-audit PASS, live challenge honest-SKIP. 10 reviewed commits; 7 real defects fixed; every change independently code-reviewed (per-wave). **-rc precedes the final store release**, whose one remaining gate is the operator-blocked **live-7187**

detect→send→torrent-in-qBittorrent verification (§11.4.21 — needs / Volumes/T7 shared into the podman VM) + operator store assets (screenshots/promo). The -rc is loadable, packaged (chrome+firefox zips), and comprehensively unit/integration/security/stress/chaos/perf/a11y/UX/challenge-tested.